ASPECT WAVE LLC

VMR: GM from Roll Period

Vessel Measurement Recorder · November 2026 · Prepared by Sophie Scopazzi

VMR data from Oct 31 and Nov 3 Nov 2026 (LT) · *Alaskan Legend* voyage 2613 (southbound leg), fully laden

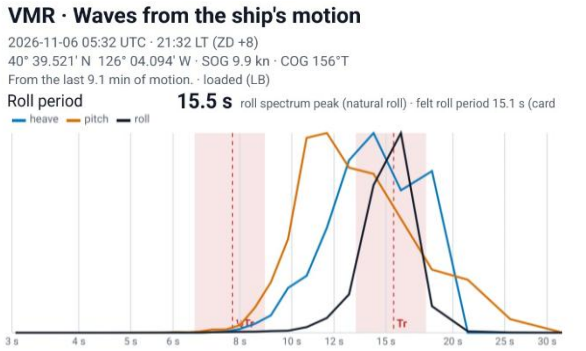
Bottom line: the IMO formula the VMR uses came within 3% of CargoMax on both days (median), with 95 % of 479 values inside ±10%. The basic formula calculated GM 52–58 % high.

The Two Formulas

Both come from the free-rolling period of a ship, where k is the roll radius of gyration:

$$T = \frac{2\pi k}{\sqrt{g \cdot GM}}$$

Both of the following formula rearrange this equation to calculate GM from a measured roll period. They only differ in how they estimate the k value.



IMO (IS Code 2008, Part A 2.3) used by the VMR

$$T = \frac{2CB}{\sqrt{GM}} \quad \text{so} \quad GM = \left(\frac{2CB}{T}\right)^2 \quad \text{with}$$
$$C = 0.373 + 0.023\,\frac{B}{d} - 0.043\,\frac{L}{100}$$

Basic

$$T = \frac{0.787B}{\sqrt{GM}} \quad \text{so} \quad GM = \left(\frac{0.787B}{T}\right)^2$$

The VMR isolates the natural roll period by calculating the roll spectrum peak, found by running an FFT on the last 17 minutes of roll data and picking the highest bin, roll_tp_s (example seen above) for T roll period, in s. B beam. d mean draft. L waterline length (LBP when entered, as here, else LOA). GM metacentric height, all in m. C roll coefficient, ≈ k/B with g = 9.81 m/s².

Because 0.787 = 2 × 0.3935, **the basic formula is the IMO formula with C fixed at 0.3935**, meaning it assumes k ≈ 0.39 B for every ship and every loaded condition. The IMO formula works C out from the hull: it rises as the ship gets wider for its draft and falls as it gets longer. For the *Alaskan Legend*, voyage 2613 MMC Closed after loading in Valdez had C = 0.317.

Why the Basic Formula Reads High

GM goes with C². For the same roll period the two always differ by the same factor, whatever the sea state:

$$\frac{GM_{basic}}{GM_{IMO}} = \left(\frac{0.3935}{C}\right)^2 = \left(\frac{0.3935}{0.317}\right)^2 = 1.54$$

A C that is 24 % too large becomes a GM that is 54 % too large. The roll period does not separate them: both have GM ∝ 1/T², so a 1 s error at 15 s moves both by about 13 %.

Results

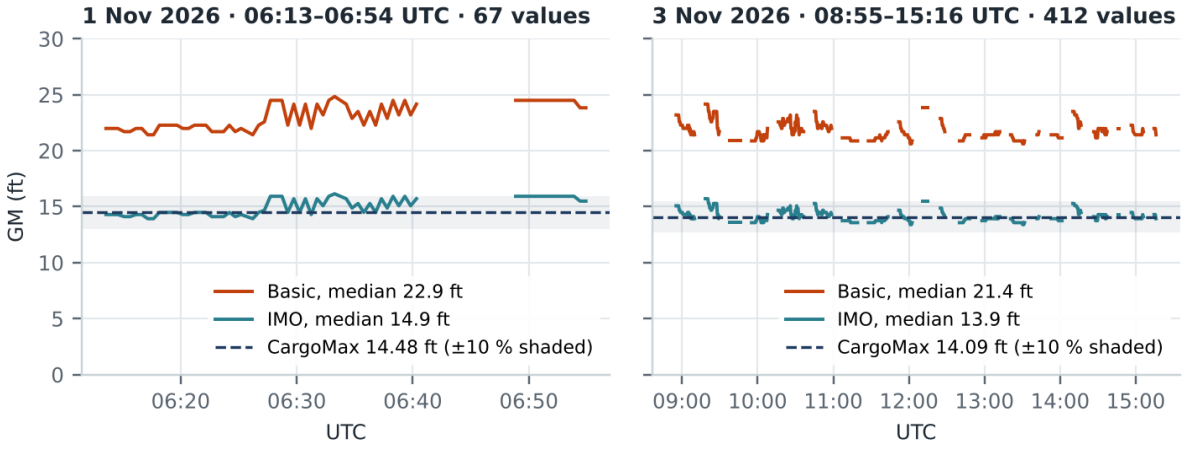


Figure 1. GM from each roll-period value with both formulas. Gaps are where the VMR logged no roll period; they are not bridged. Each calculated basic value is 1.54 × the IMO value at the same time. Ship observing ZD +8.

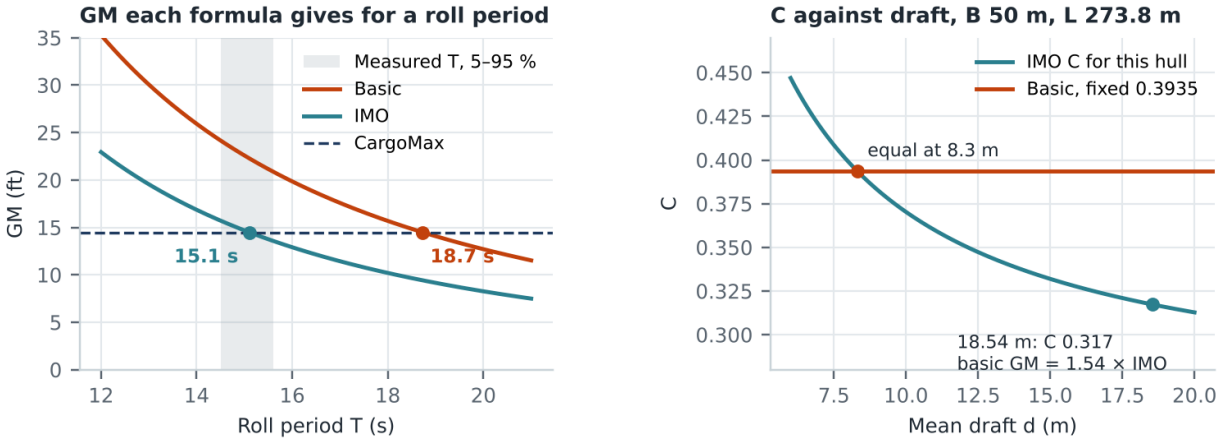


Figure 2. Left: GM against roll period, with the range the ship rolled at. Right: IMO C for this hull against draft.

Day (values)	Median T	CargoMax GM	IMO GM	IMO error	IMO within ±10 %	Basic GM	Basic error	Basic within ±10 %
1 Nov (67)	14.9 s	14.48 ft	14.9 ft	+3 %	72 %	22.9 ft	+58 %	0 %
3 Nov (412)	15.4 s	14.09 ft	13.9 ft	−1 %	99 %	21.4 ft	+52 %	0 %
Both (479)	**15.3 s**	**14.14 ft**	**14.1 ft**	**0 %**	**95 %**	**21.7 ft**	**+54 %**	**0 %**

Table 1. Medians against CargoMax 14.48 ft (Nov 1) and 14.09 ft (3 Nov). Error = (median − CargoMax) / CargoMax; for Both, the median of each value's error against its own day's CargoMax. "Within ±10 %" counts every value.

Conclusion

- **The IMO formula is a more accurate calculation of a vessel's GM.** It allows for the hull and its draft; the basic formula assumes one shape for every ship.
- **The Vessel Measurement Recorder** estimated the *Alaskan Legend*'s fully laden GM within 3% of the CargoMax calculated GM value.

Appendix: Where is k in the roll period formula?

> **In short.** Neither formula drops k. Both fold it into the coefficient C, which is the roll radius of gyration as a fraction of the beam (C ≈ k/B). The basic formula fixes it; the IMO formula estimates it from the hull's proportions.

The Starting Formula

A ship rolling freely at her natural period behaves like a pendulum. Its period is set by the roll radius of gyration k and the metacentric height:

$$T = \frac{2\pi k}{\sqrt{g \cdot GM}}$$

The above equation is standard naval architecture, not something the IS Code states. MSC.267(85) (the 2008 IS Code) only gives the approximate form T = 2CB/√GM, with no derivation.

Folding k into C

Write k as a fraction of the beam, k = (k/B)·B, and regroup:

$$T = \frac{2\pi}{\sqrt{g}} \cdot \frac{k}{B} \cdot \frac{B}{\sqrt{GM}}$$

The IMO formula has the form T = 2CB/√GM. Setting the two equal, B and √GM cancel:

$$C = \frac{\pi}{\sqrt{g}} \cdot \frac{k}{B} = 1.003\,\frac{k}{B} \quad \text{so} \quad k \approx C \cdot B$$

With g = 9.81 m/s², π/√g = 1.003, so C and k/B are effectively the same number.

Why k Has to Be Estimated

The k that sets the roll period is not just the ship's own mass distribution. It also includes the water the hull carries with it as it rolls (added mass), so it cannot be worked out from the loading computer's weights. Both formulas therefore rely on a fitted value. The IS Code does not say where its C formula came from.

Sources

- IMO Resolution MSC.267(85), International Code on Intact Stability, 2008, Part A 2.3.4
- Vassalos, Jasionowski & Cichowicz, "Weather Criterion – Questions and Answers", STAB 2003

2.3.4 The angle of roll (φ_1)[8] referred to in 2.3.1.2 shall be calculated as follows:

$$\varphi_1 = 109 * k * X_1 * X_2 * \sqrt{r * s} \quad (degrees)$$

where:

X_1 = factor as shown in table 2.3.4-1

X_2 = factor as shown in table 2.3.4-2

k = factor as follows:

k = 1.0 for round-bilged ship having no bilge or bar keels

k = 0.7 for a ship having sharp bilges

k = as shown in table 2.3.4-3 for a ship having bilge keels, a bar keel or both

r = $0.73 + 0.6\ OG/d$

This k value is a bilge keel factor; different than then k value used throughout this document

with:

OG = KG – d

d = mean moulded draught of the ship (m)

s = factor as shown in table 2.3.4-4, where T is the ship roll natural period. In absence of sufficient information, the following approximate formula can be used:

Rolling period $$T = \frac{2 * C * B}{\sqrt{GM}} (s)$$

where:

$$C = 0.373 + 0.023(B/d) - 0.043(L_{wl}/100).$$

The symbols in tables 2.3.4-1, 2.3.4-2, 2.3.4-3 and 2.3.4-4 and the formula for the rolling period are defined as follows:

L_{wl} = length of the ship at waterline (m)

B = moulded breadth of the ship (m)

d = mean moulded draught of the ship (m)

8 The angle of roll for ships with anti-rolling devices should be determined without taking into account the operation of these devices unless the Administration is satisfied with the proof that the devices are effective even with sudden shutdown of their supplied power.

C_B = block coefficient (-)

A_k = total overall area of bilge keels, or area of the lateral projection of the bar keel, or sum of these areas (m²)

GM = metacentric height corrected for free surface effect (m).

Tables not used or required

Table 2.3.4-1 – Values of factor X_1

B/d	X_1
≤ 2.4	1.0
2.5	0.98
2.6	0.96
2.7	0.95
2.8	0.93
2.9	0.91
3.0	0.90
3.1	0.88
3.2	0.86
3.4	0.82
≥ 3.5	0.80

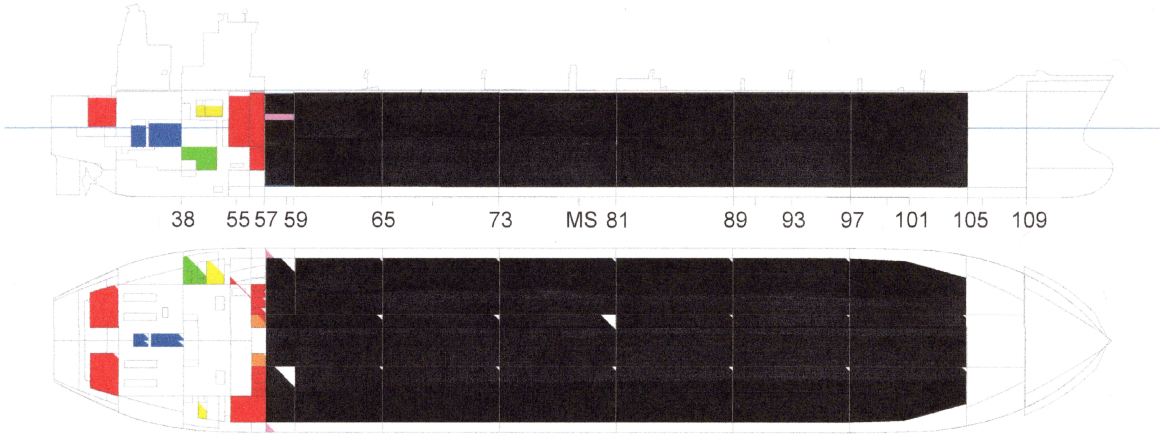
Table 2.3.4-2 – Values of factor X_2

C_B	X_2
≤ 0.45	0.75
0.50	0.82
0.55	0.89
0.60	0.95
0.65	0.97
≥ 0.70	1.00

Table 2.3.4-3 – Values of factor k

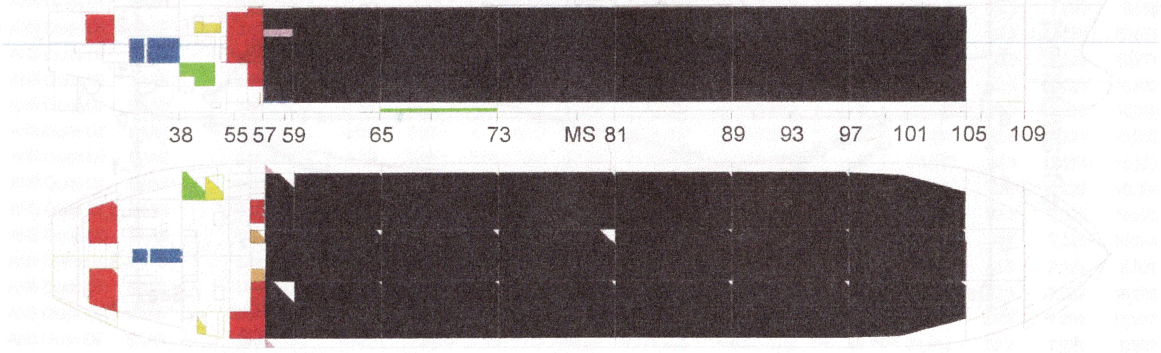
$\frac{A_k \times 100}{L_{WL} \times B}$	k
0	1.0
1.0	0.98
1.5	0.95
2.0	0.88
2.5	0.79
3.0	0.74
3.5	0.72
≥ 4.0	0.70

MMC Closed
Trim and Stability Summary



Loading					
Displacement	221,194	MT	Specific Gravity	1.025	
Deadweight	182,173	MT			
Avail Deadweight	10,875	MT	--190K DWT--		
Stability			**Trim**		
KMt	67.70	ft	LCF Draft	61.56	ft
VCG (Upright)	51.20	ft	LCF	432.63	Ft-AP
GMt (Solid)	16.50	ft	LCB	460.37	Ft-AP
FS Correction	2.02	ft	LCG	461.01	Ft-AP
GMt (Corrected)	14.48	ft	TP1in	320.7	LT/in
Intact GMt Required	11.90	ft	MT1in	20,788	ft-LT/in
Intact GMt Margin	2.58	ft	Trim at Perps	-0.56	ft
--Combined Intact Required GM--			Heel Angle	0.04S	deg
			Propeller Immersion	244.05	%
Dam Stab Assesment	Needs Update				
Damage Stability compliance has not been demonstrated.					
Please run the direct damage assessment.					
Drafts - Perps			**Drafts - Perps w/Deflections**		
AP	61.29 (61-3.5)	ft	AP	60.89 (60-10.6)	ft
MS	61.57 (61-6.9)	ft	MS	61.98 (61-11.7)	ft
FP	61.85 (61-10.2)	ft	FP	61.45 (61-5.4)	ft
Drafts - Marks			**Drafts - Marks w/Deflections**		
Aft	61.29 (61-3.5)	ft	Aft	60.87 (60-10.4)	ft
MS	61.57 (61-6.9)	ft	MS	61.98 (61-11.7)	ft
Fwd	61.85 (61-10.2)	ft	Fwd	61.46 (61-5.6)	ft
Strength --At Sea--					
Shear (Min)	-11,699	MT	171.33 Ft-AP	73.12	%
Shear (Max)	11,809	MT	794.95 Ft-AP	59.54	%
Shear (Max %Allow)	-11,699	MT	171.33 Ft-AP	73.12	%
Moment (Max Hog)	---	ft-LT	--- Ft-AP	---	%
Moment (Max Sag)	2,436,590S	ft-LT	481.78 Ft-AP	---	%
Moment (Max %Allow)	2,436,536S	ft-LT	483.14 Ft-AP	91.18	%
Deflection	9.7S	in	457.15 Ft-AP		
Bridge Visibility - ABS					
Actual	862.64	ft	Required	<1,640.42	ft
Notes					
Drafts from Keel					
Hull from Tables					
Tanks from Tables					

Daily Soundings
Trim and Stability Summary



Loading					
Displacement	221,437	MT	Specific Gravity	1.025	
Deadweight	182,416	MT			
Avail Deadweight	10,632	MT	--190K DWT--		
Stability			**Trim**		
KMt	67.69	ft	LCF Draft	61.62	ft
VCG (Upright)	51.15	ft	LCF	432.62	Ft-AP
GMt (Solid)	16.55	ft	LCB	460.33	Ft-AP
FS Correction	2.46	ft	LCG	460.99	Ft-AP
GMt (Corrected)	14.09	ft	TP1in	320.7	LT/in
Intact GMt Required	11.89	ft	MT1in	20,793	ft-LT/in
Intact GMt Margin	2.19	ft	Trim at Perps	-0.57	ft
--Combined Intact Required GM--			Heel Angle	0.03S	deg
			Propeller Immersion	244.28	%
Dam Stab Assesment	Needs Update				
Damage Stability compliance has not been demonstrated.					
Please run the direct damage assessment.					
Drafts - Perps			***Drafts - Perps w/Deflections***		
AP	61.35 (61-4.2)	ft	AP	60.94 (60-11.3)	ft
MS	61.63 (61-7.6)	ft	MS	62.04 (62-0.5)	ft
FP	61.92 (61-11.1)	ft	FP	61.51 (61-6.2)	ft
Drafts - Marks			**Drafts - Marks w/Deflections**		
Aft	61.34 (61-4.1)	ft	Aft	60.92 (60-11.1)	ft
MS	61.63 (61-7.6)	ft	MS	62.04 (62-0.5)	ft
Fwd	61.92 (61-11.0)	ft	Fwd	61.53 (61-6.4)	ft
Strength --At Sea--					
Shear (Min)	-11,893	MT	171.33 Ft-AP	74.33	%
Shear (Max)	11,817	MT	794.95 Ft-AP	59.58	%
Shear (Max %Allow)	-11,893	MT	171.33 Ft-AP	74.33	%
Moment (Max Hog)	---	ft-LT	--- Ft-AP	---	%
Moment (Max Sag)	2,453,204S	ft-LT	480.61 Ft-AP	---	%
Moment (Max %Allow)	2,453,019S	ft-LT	483.14 Ft-AP	91.80	%
Deflection	9.8S	in	457.15 Ft-AP		
Bridge Visibility - ABS					
Actual	861.16	ft	Required	<1,640.42	ft

Notes

Drafts from Keel
Hull from Tables
Tanks from Tables